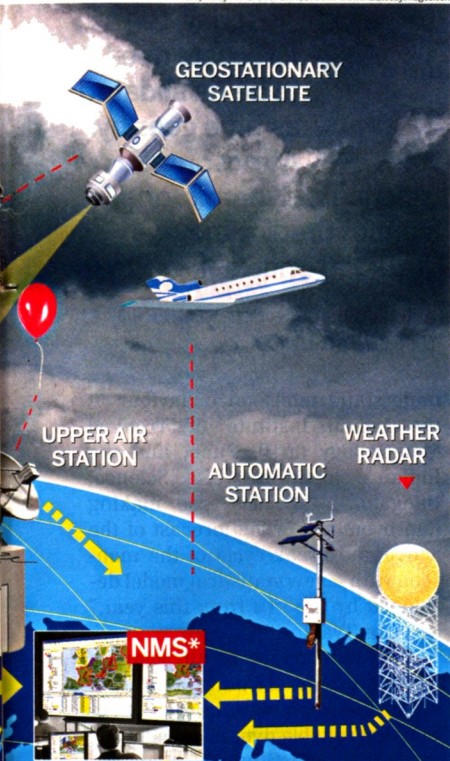




*Network Management Station

started just after April last year, and its strength peaked in November-December 2009, at a moderate level and since then there has been a decline in strength. A recent advisory of the Climate Prediction Centre of the US shows a definite breakdown of the El Nino conditions in the next couple of months. By July-end, it is likely to be in neutral condition which augurs well for the Indian monsoon. But that is a theoretical perception.

As of now, the IMD relies on the statistical ensemble forecasting system with five parameters, including data up to March for the April forecast and six parameters, including data up to May for the forecast update in June. The North Atlantic Sea Surface Temperature, Equatorial SE Indian Ocean Sea Surface Temperature, East Asia Mean Sea Level Pressure, NW Europe Land Surface Air Temperature and Equatorial Pacific Warm Water Volume are the parameters for the first forecast. Later, the

Point

Where the country's weather office is going wrong

- Instead of dynamic models, it relies on statistical ones, which have inherent limitations
- It relies excessively on historical data that is significantly related to the total rainfall received by the country as a whole during June-September period
- Predictive signals observed prior to the monsoon season are assumed to persist during the season
- It cannot factor out sudden changes occurring in the global large scale features such as the El Nino that can have a huge impact on the monsoon rainfall
- The climatology or spatial distribution of the rain during the season cannot be simulated



“IMD forecasts of the active and break cycles are still evolving. For this to happen a national mission on monsoon is on the cards.”

AIR VICE-MARSHAL (RETD) AJIT TYAGI, DG, IMD



“Infrastructure, including weather stations and radars, needs to be replaced or upgraded because of technological obsolescence.”

VISHAL SINGH, COUNTRY MANAGER, FORECA

first three of these parameters along with Central Pacific Sea Surface Temperature Tendency, North Atlantic Mean Sea Level Pressure and North Central Pacific Wind at 1.5 km above sea level are used to make the six-parameter forecast. But these statistical ensembles introduced in 2007 are yet to establish their credibility as these have failed to signal the monsoon accurately in two of the three years of use.

The IMD often latches on to one or the other physical phenomenon to make a prediction. These isolated forecasts have triggered undeserved excitement. The latest is on when the monsoon will break over India. “It may hit the Kerala coast a week to 10 days ahead of schedule,” says P.V. Joseph, former director, IMD, and now professor emeritus at the Department of Atmospheric Science at the Cochin University of Science and Technology.

His forecast is based on a phenomenon known as pre-monsoon rain peak in which the temperature of the Bay of Bengal off the east coast rises about 40 days before the monsoon and clouds near the equator move north to bring rains on the southern tip of India. “It is an indicator of coming monsoons and in most years it has come true,” says Joseph. On the other hand, researcher Andy Turner at Reading University, UK, has found how snow conditions over the Himalayas and Tibetan plateau could be used to help forecast seasonal monsoon during the onset month of June. The latest research shows that greater snowfall reflects more sunlight and produces a cooling over the Himalayas which, in turn, will weaken monsoon winds that bring rain.

Sophisticated data-gathering tools like satellite imageries, radars, automatic weather stations, GPS Sonde and the Global Telecom System have made it possible for the IMD to analyse information, including global surface and upper air data from high resolution regional and global numerical models, to offer lucid interpretations. This has also enabled it to issue forecasts in different time frames: short range